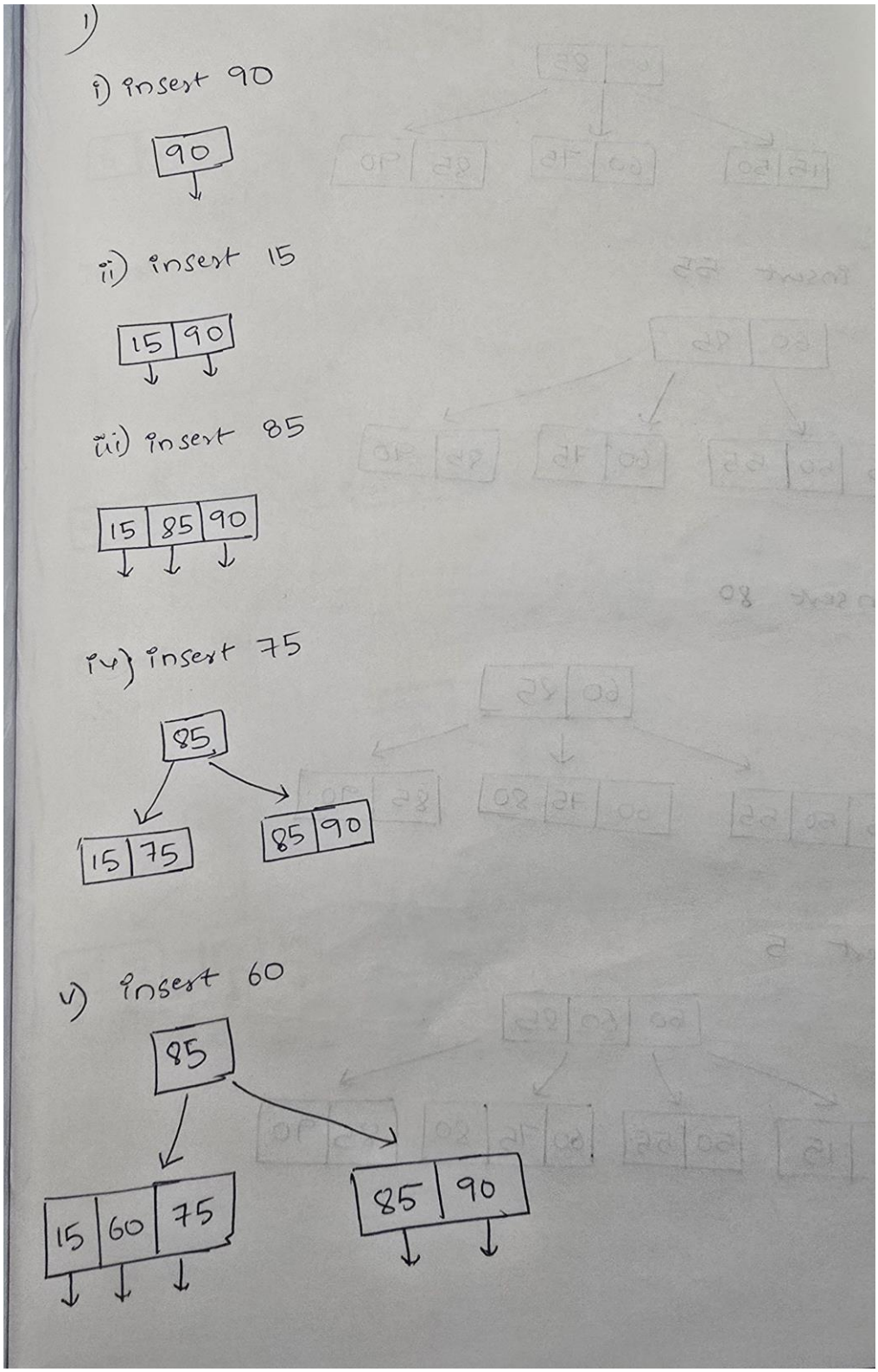
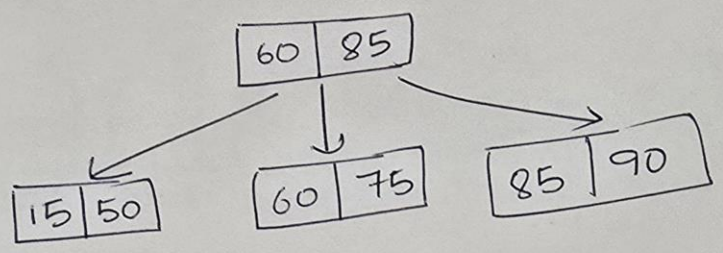
I Bharath Chandra Nallani declare that I have completed this assignment completely and entirely on my own, without any consultation with others. I understand that any breach of the UAB Academic Honor Code may result in severe penalties.

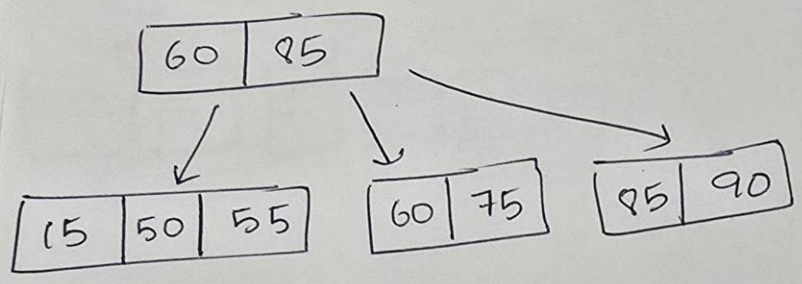
1. B+ tree:



insert 50



insert 55



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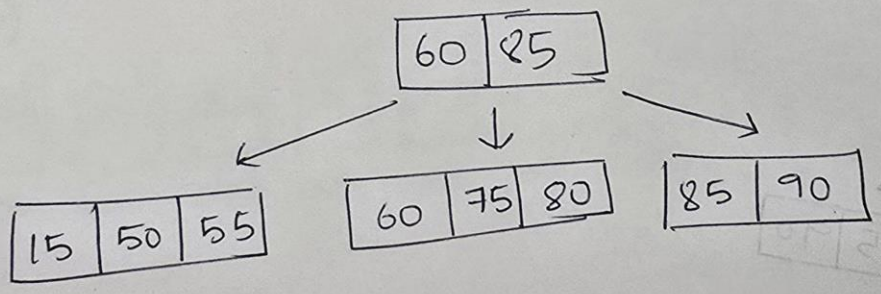

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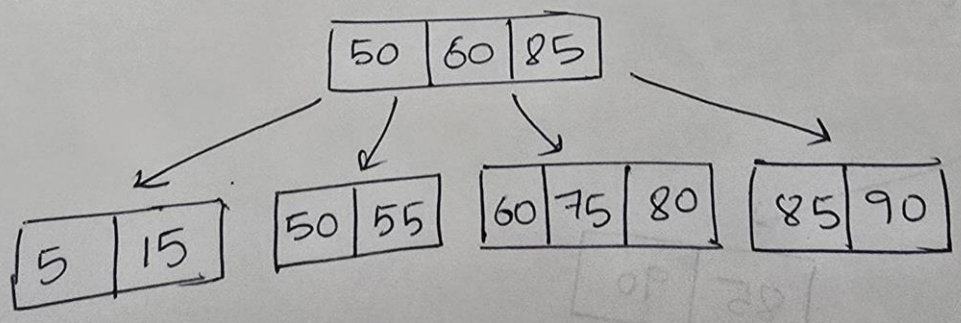

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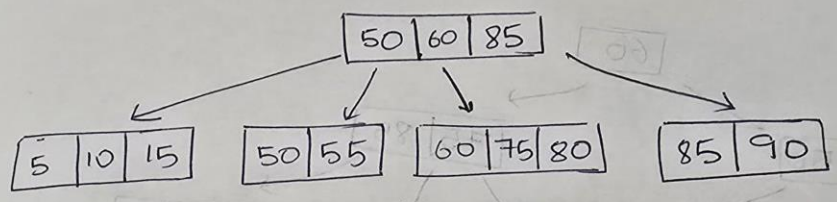
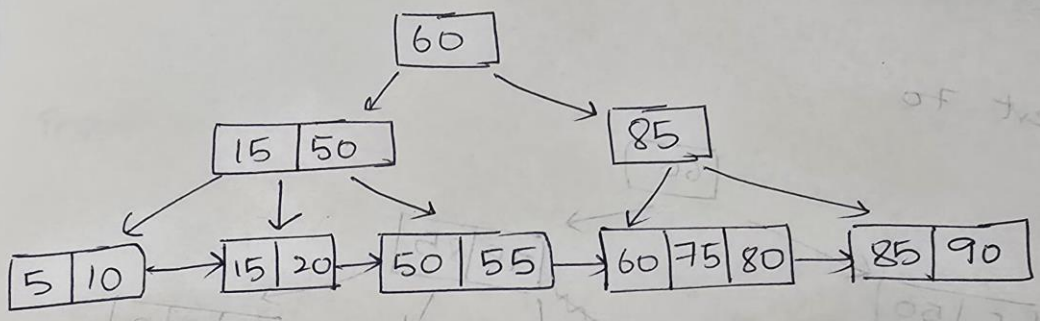
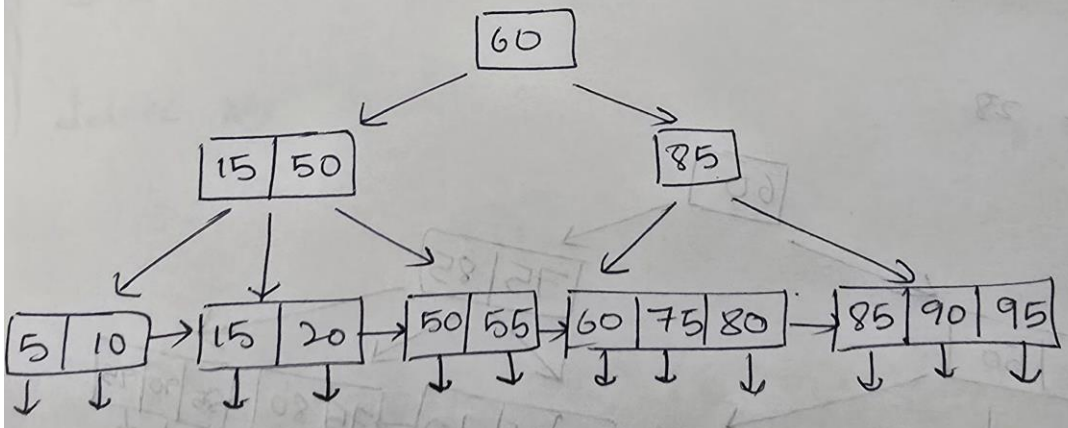


insert 20



insert - 95



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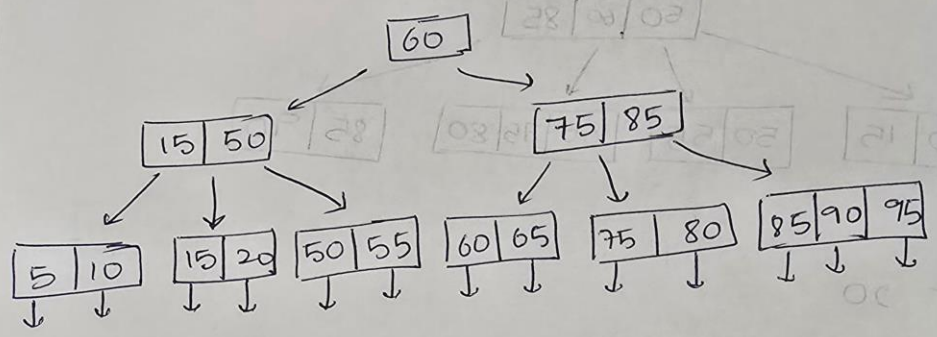

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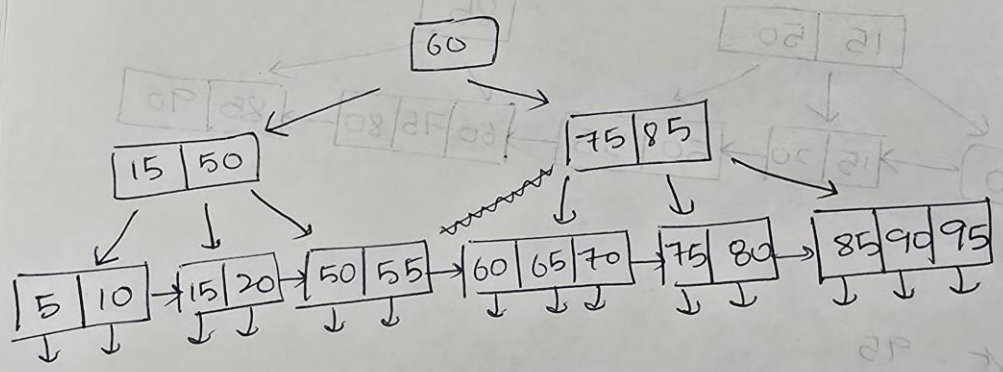

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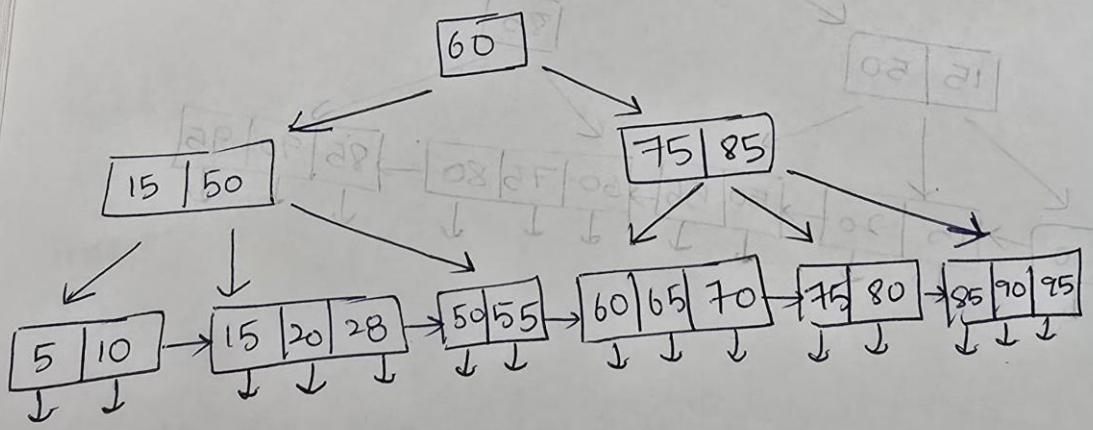
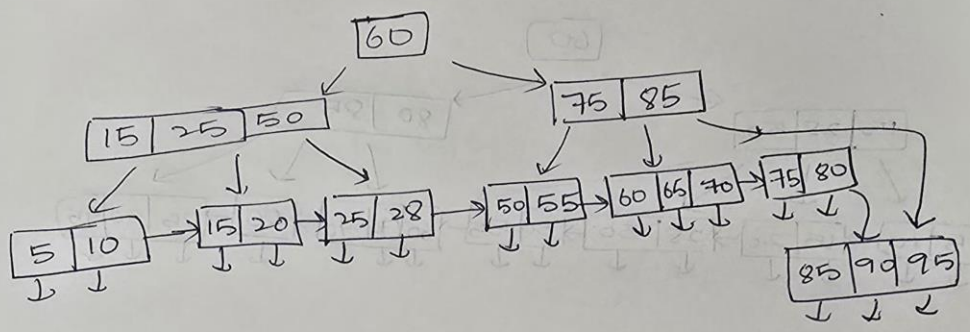
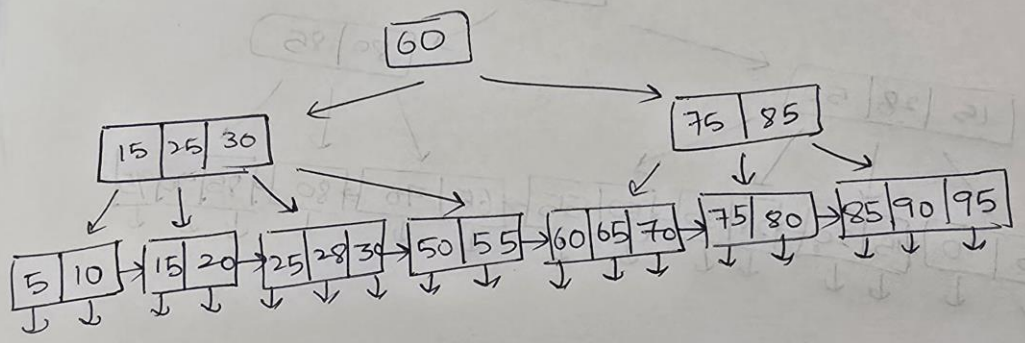


Insert 25



Insert 30



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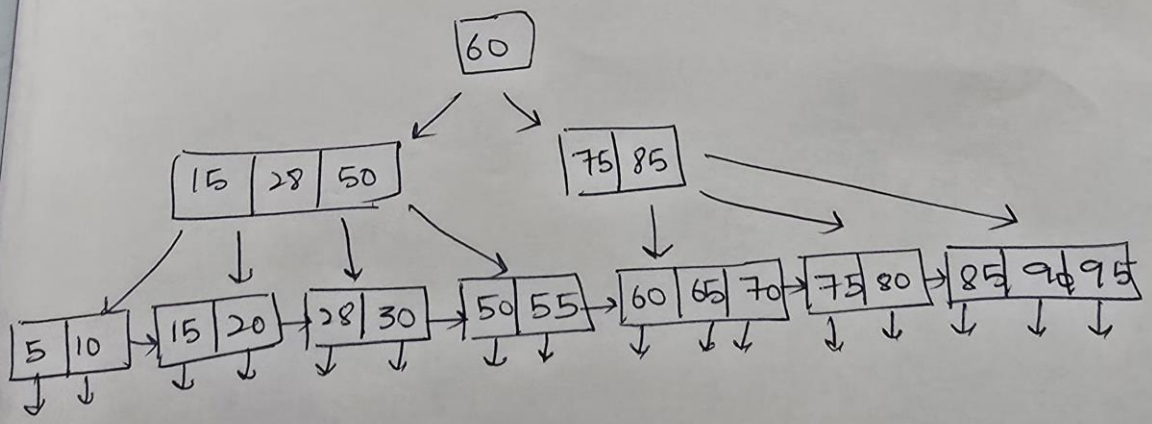

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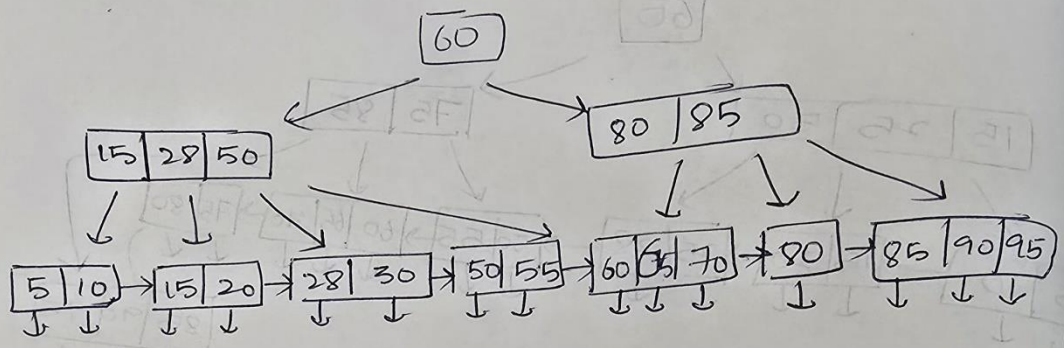
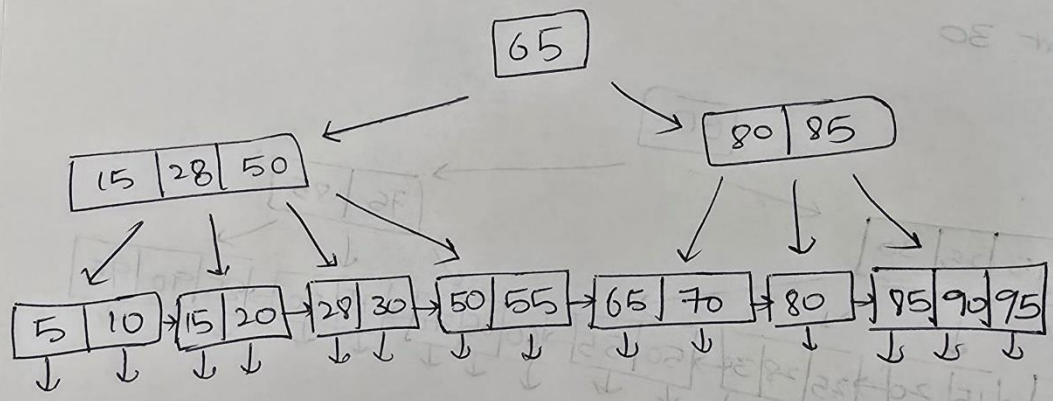
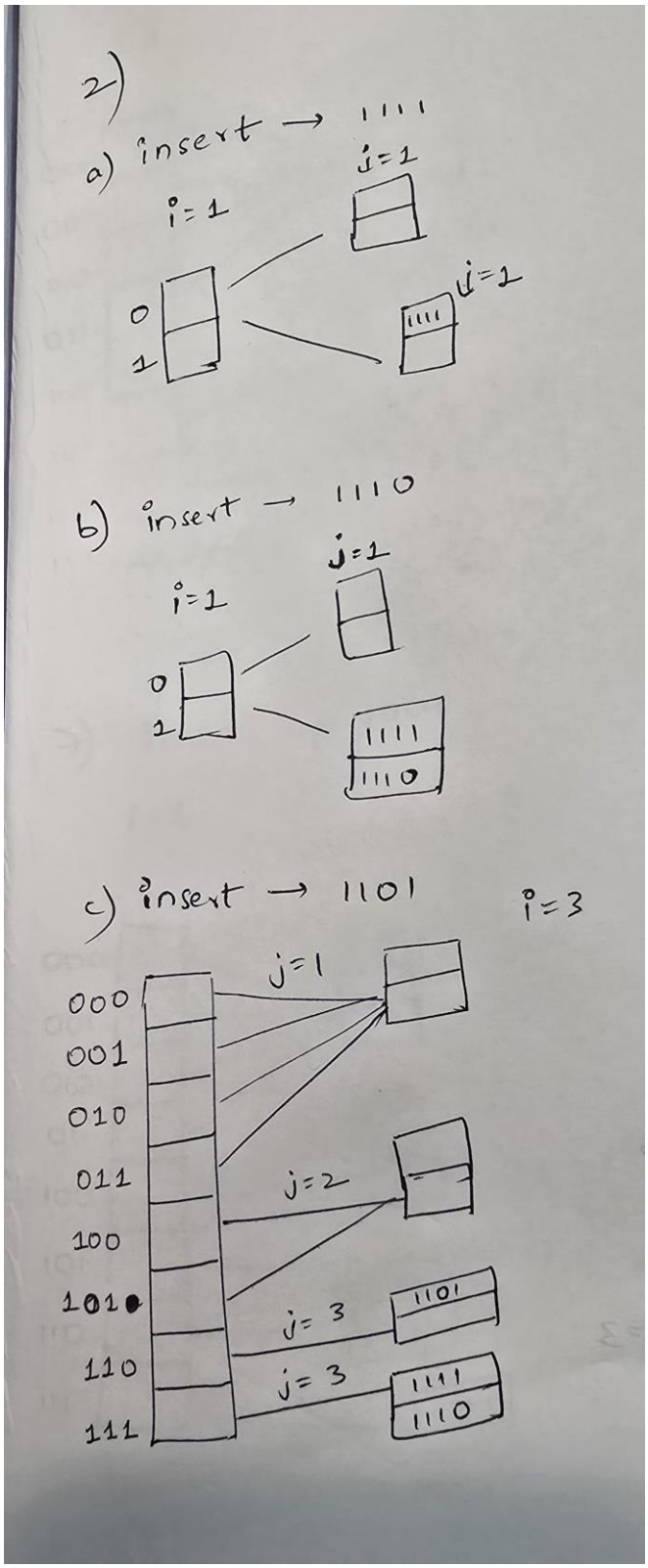


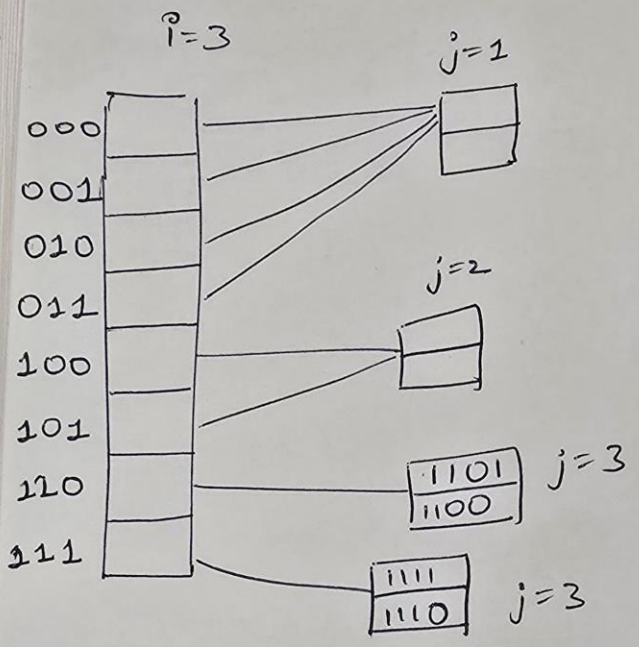
delete 60:



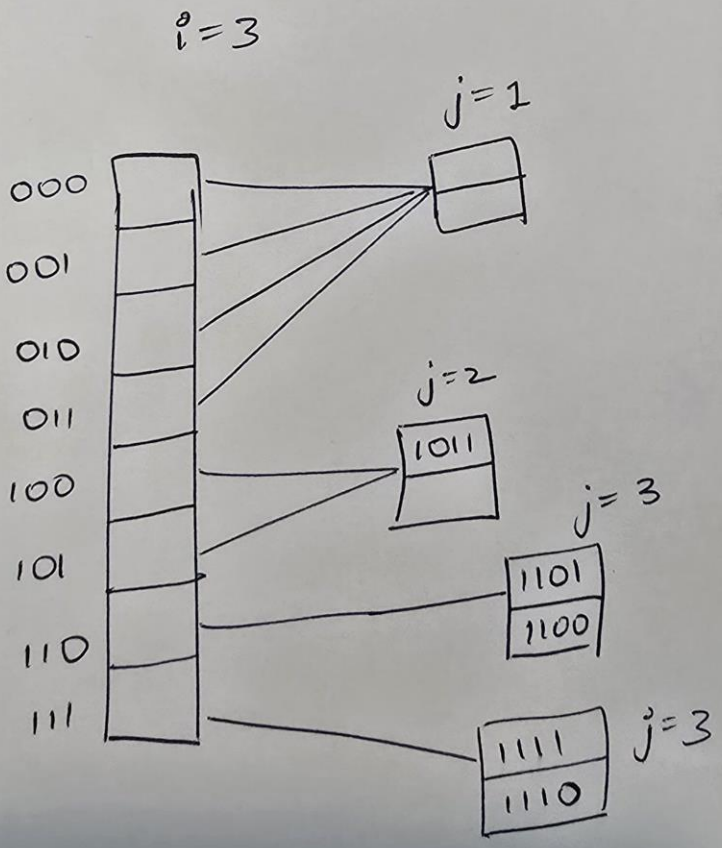
2. **Extensible Hash Table:**



4 d) insert → 1100

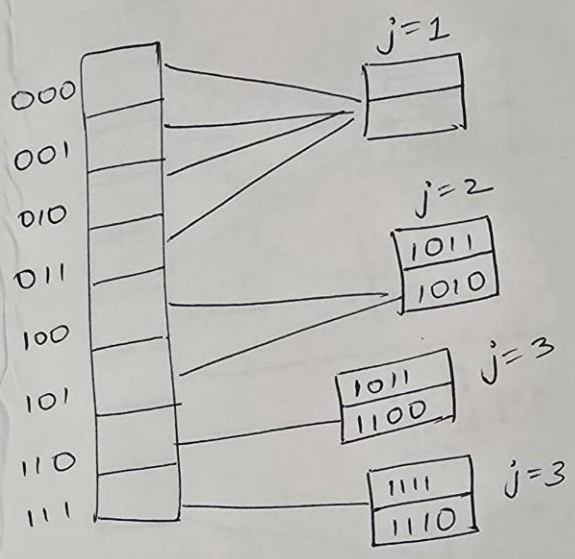


5) insert → 1011



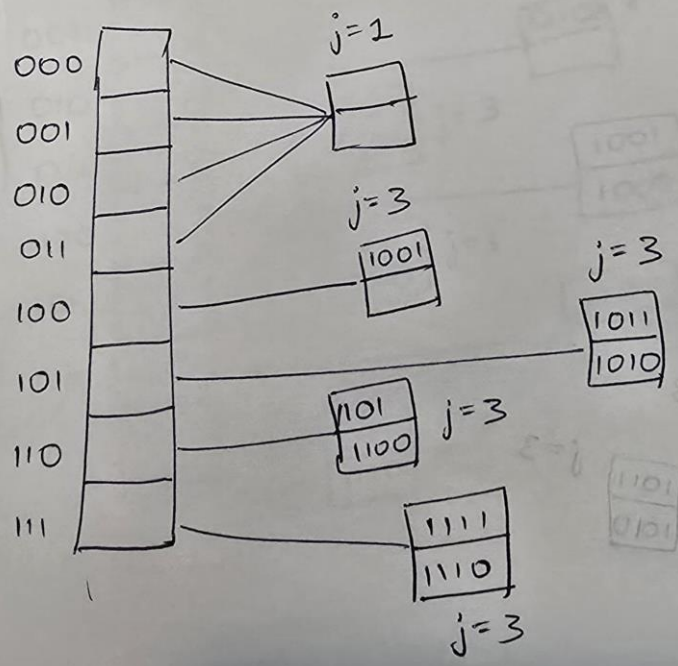
6) Insert → 1010

$i=3$

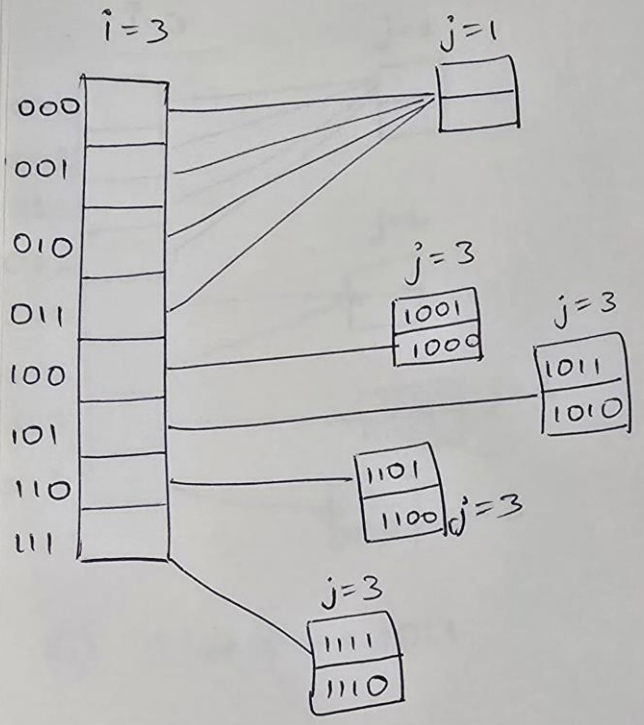


7) Insert → 1001

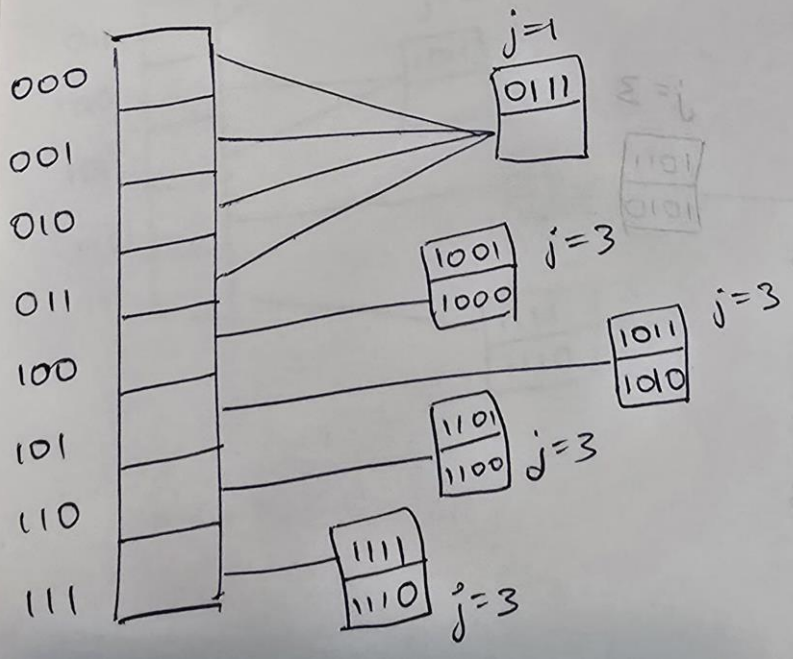
$i=3$



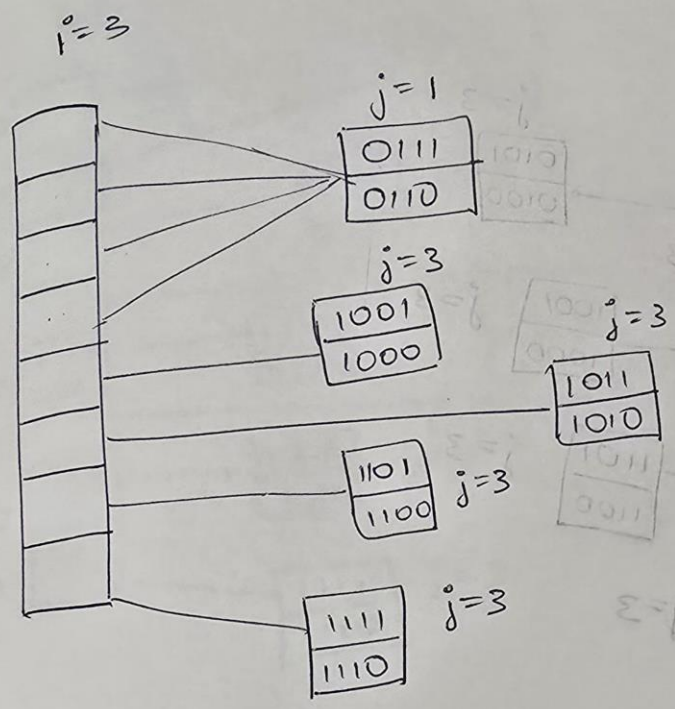
8) insert → 1000



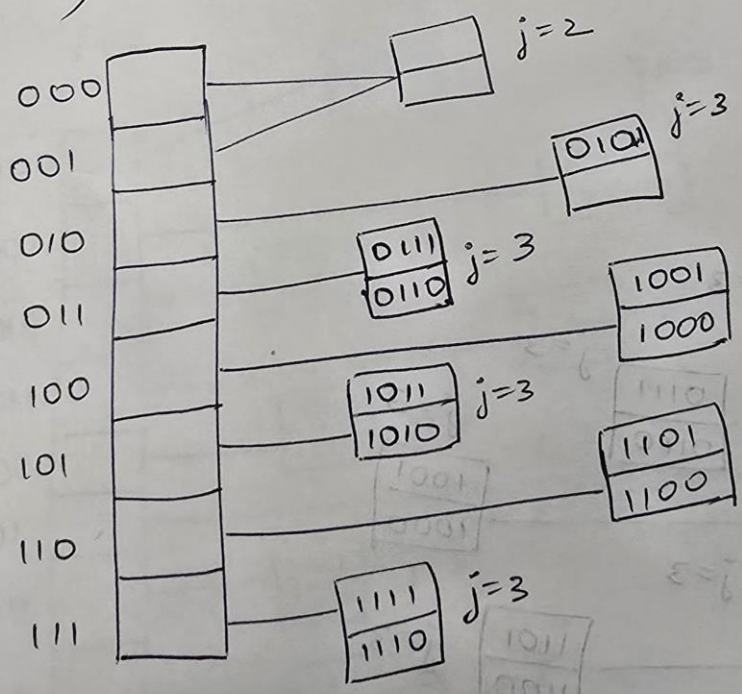
9) insert → 0111



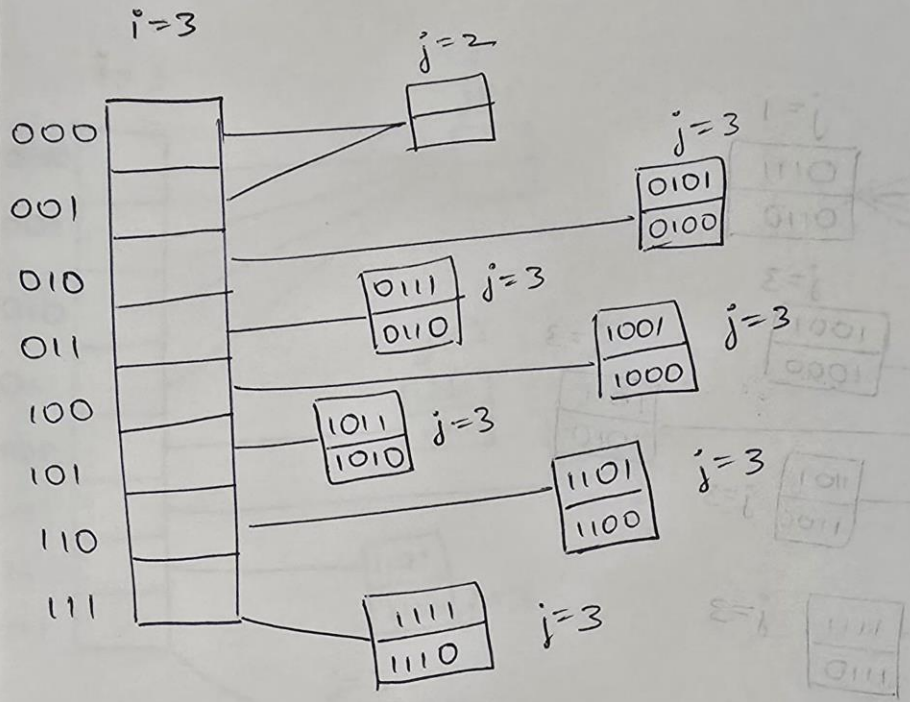
10) insert → 0110



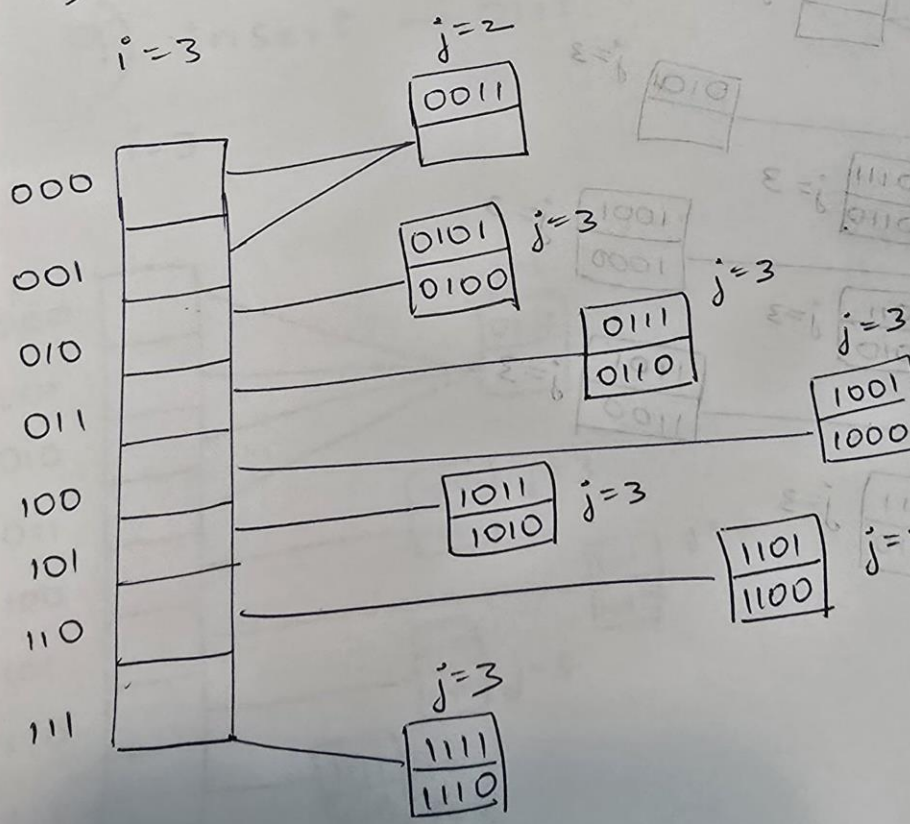
11) insert → 0101



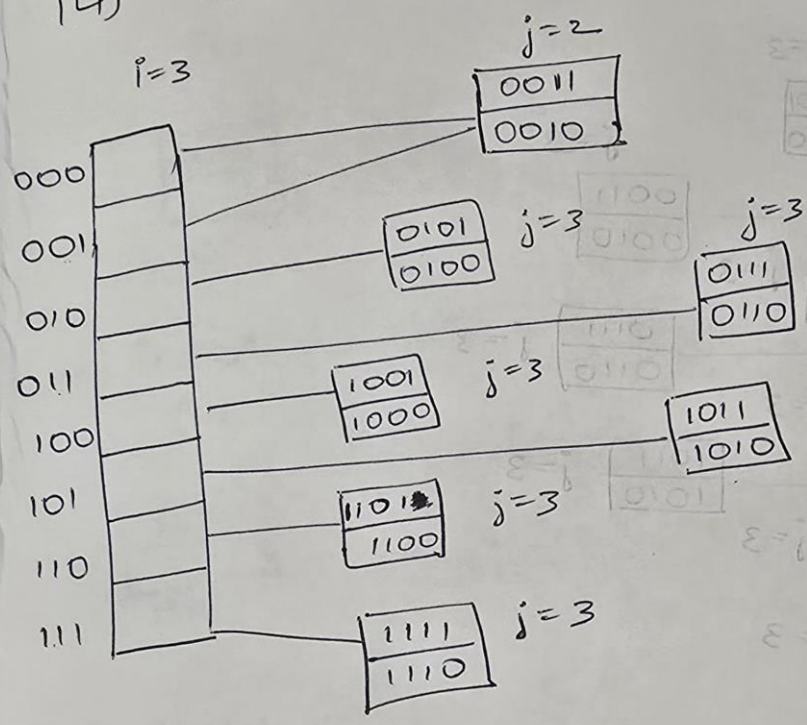
12) insert → 0100



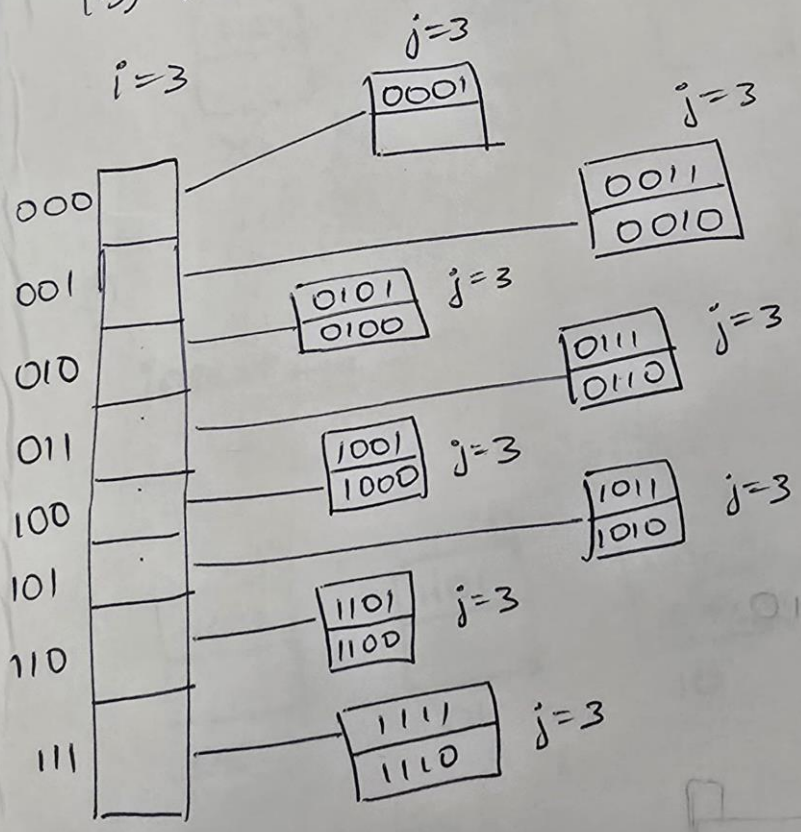
13) insert → 0011

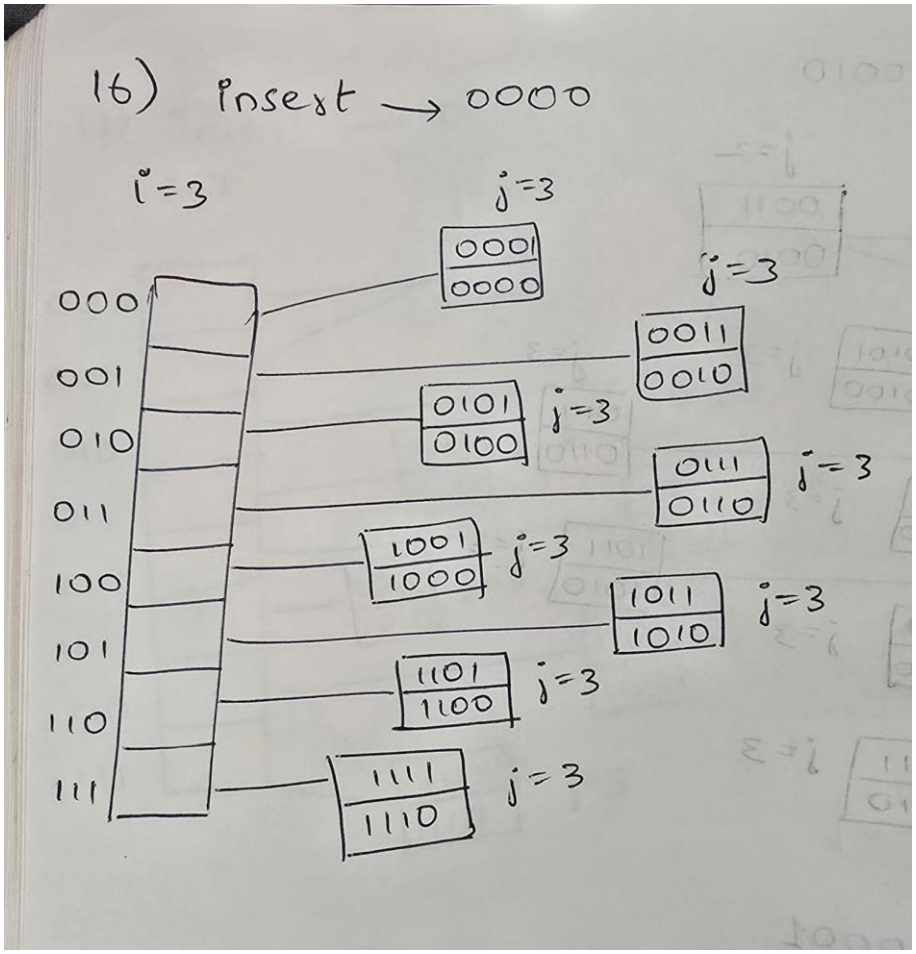


14) insert → 0010



15) insert → 0001



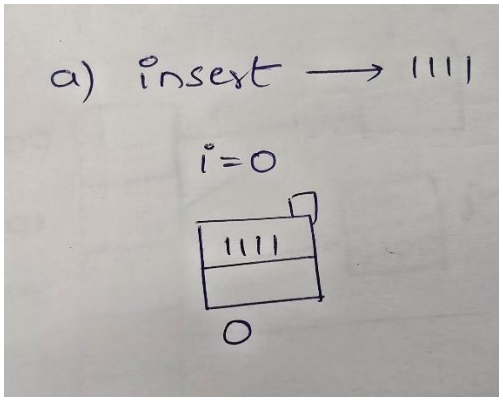


3. Linear Hash Table:

a. i = 0

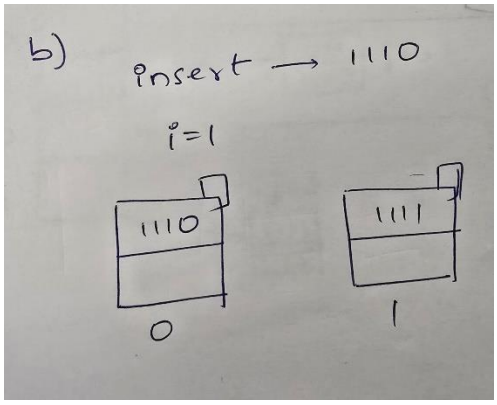
U = 1/1 = 1 which is < 1.5

Therefore, not any increase in size of bucket.



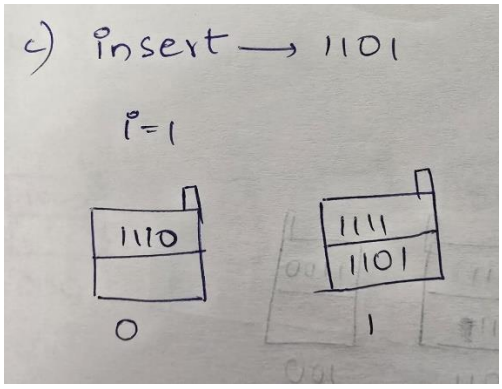
b) U= 2/1 = 2
which is greater than 1.5

Therefore, increase in size of bucket and i =1



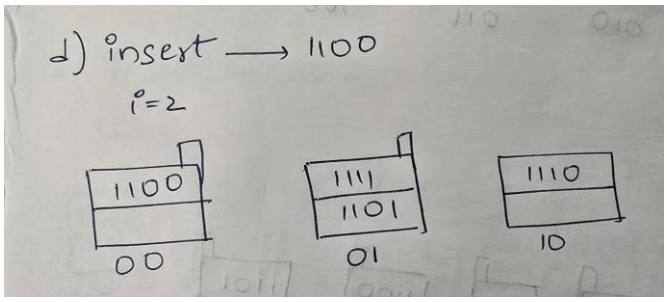
c)

U=3 / 2 = 1.5
Therefore, no increase in size of bucket



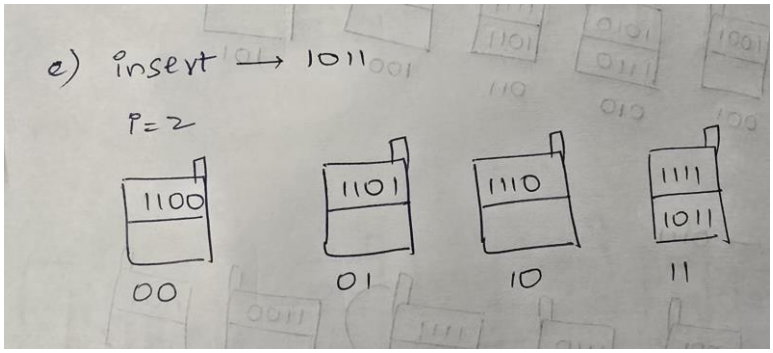
d)

U=4 / 2 = 2
greater than 1.5.
Therefore, increase in size of bucket
and i =2



e)

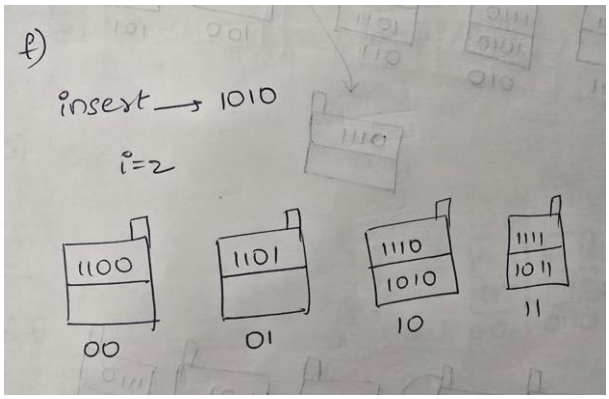
U= 5 / 3= 1.66
Greater than 1.5
Therefore, increase in size of bucket



f)

U= 6 / 4 = 1.5

Therefore, no increase in size of bucket



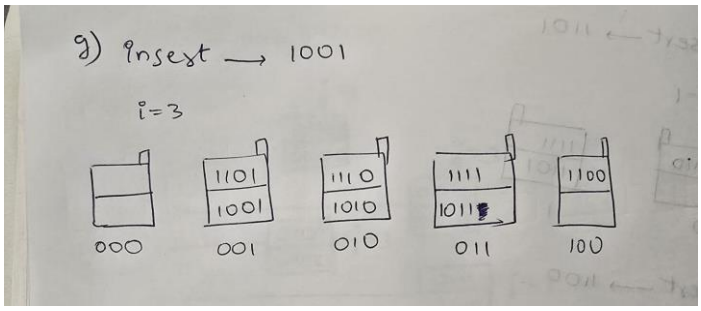
g)

U=7 / 4 = 1.75

Greater than 1.5

Therefore, increase in the size of bucket

i=3

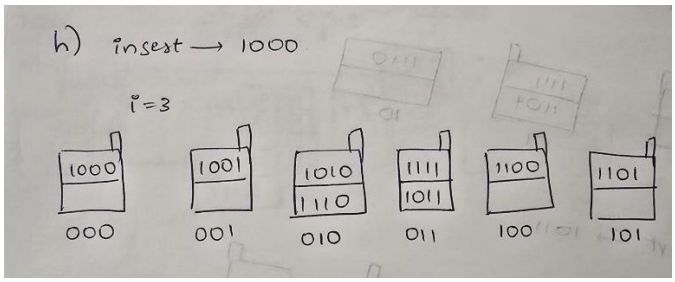


h)

U= 8 / 5 = 1.6

Greater than 1.5

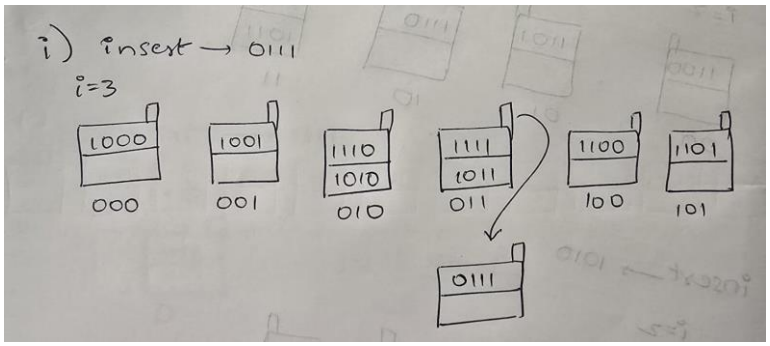
Therefore, increase in size of bucket



i)

U=9 / 6 = 1.5

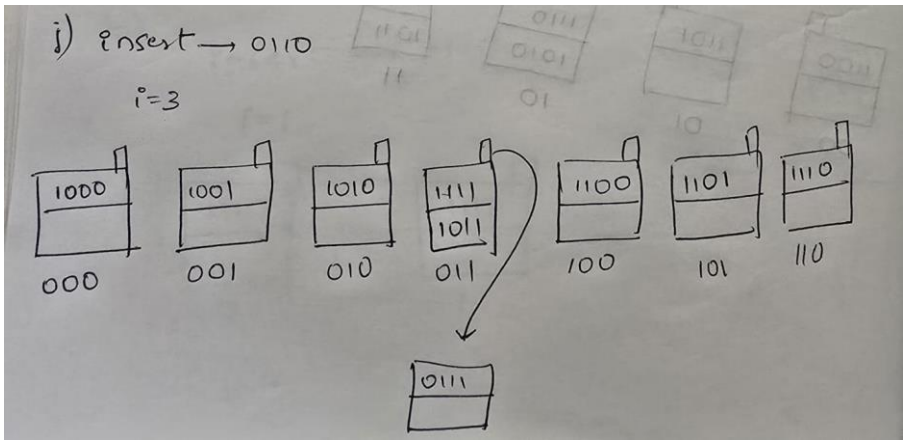
Therefore, No increase in size of bucket



j)

U= 10 / 6 = 1.66

Greater than 1.5
Therefore, increase in size of bucket

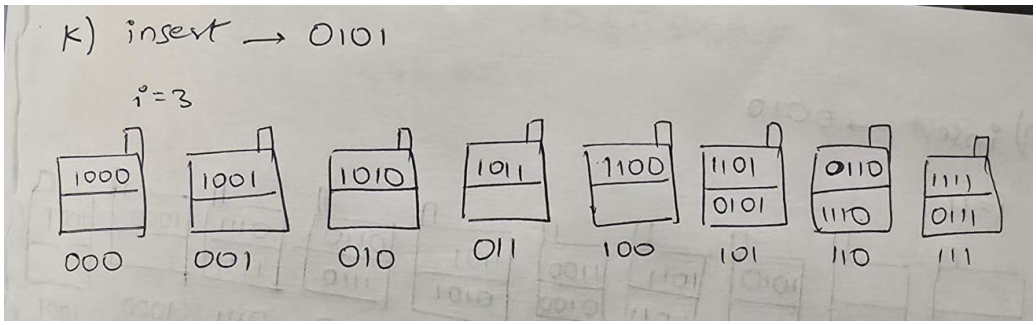


k)

U=11 / 7 =1.57

Greater than 1.5.

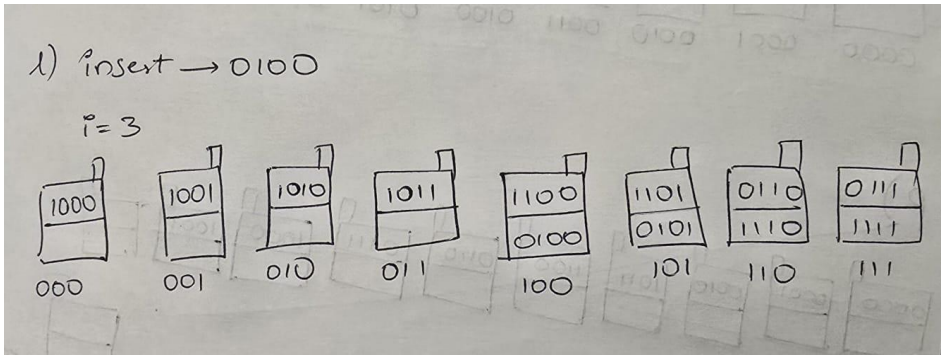
Therefore, increase in size of bucket



L)

U=12 / 8 = 1.5

Therefore, No increase in size of bucket

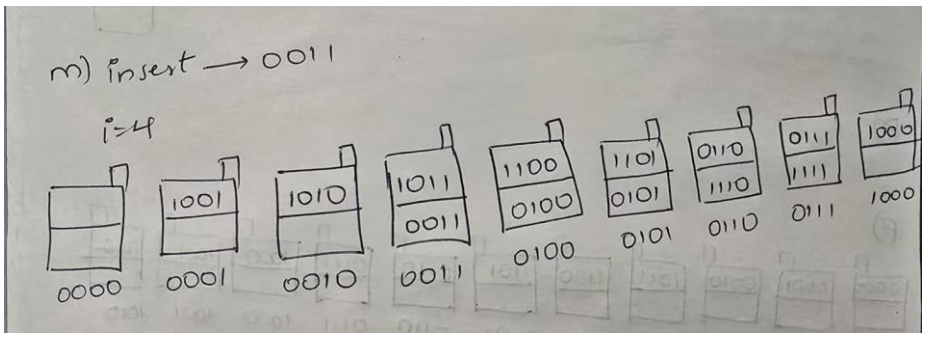


m)

U= 13 / 8 = 1.62

Greater than 1.5 increase

So, in size of bucket and i= 4

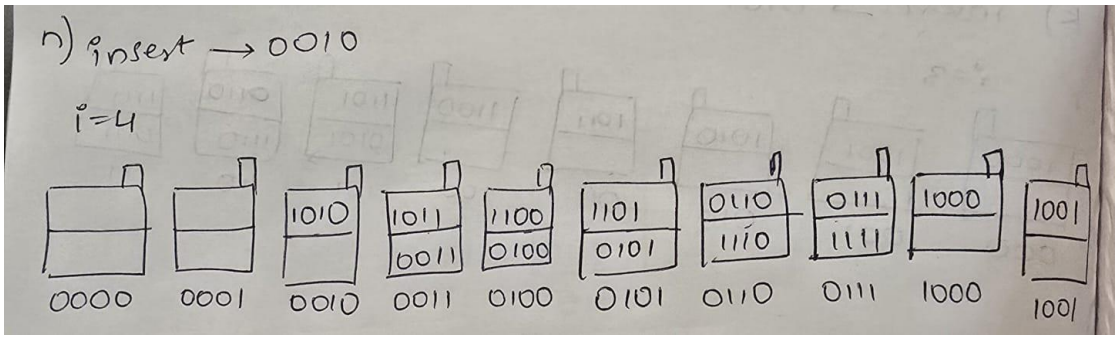


n)

U= 14 / 9 = 1.55

Greater than 1.5

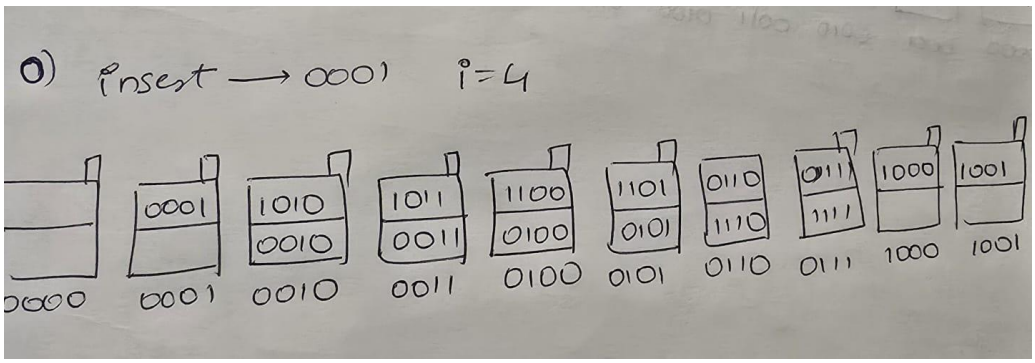
So, increase in size of bucket



O)

U= 15 / 10 = 1.5

So, no increase in size of bucket

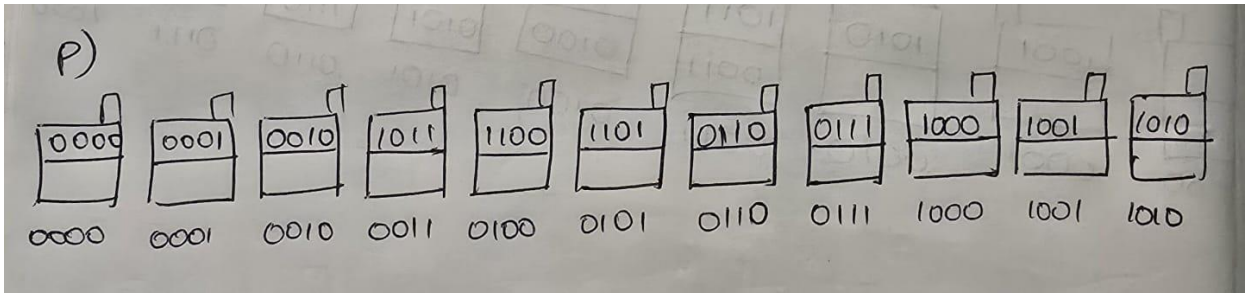


p)

U= 16 / 10 = 1.6

Greater than 1.5

So, increase in size of bucket



4. Index 1 + Dense:

Each index entry consists of:

Employee ID length: 40 bytes

Record pointer length: 8 bytes

Total per entry: 48 bytes

Available space per block: 8192 bytes – 100 bytes (for overhead) = **8092 bytes**

Number of index entries per block:
8092 / 48 ≈ **168 entries per block**

Total number of tuples: 500

Required number of blocks:
500 / 168 ≈ **2.98**, which rounds up to **3 blocks**

Index 1 + Sparse:

Index entry size: 40 bytes (emp.id) + 6 bytes (record pointer) = **46 bytes**.

Index entries per block: (8192 - 100) ÷ 46 = **176 entries/block**.

Employee data per block: (8192 - 100) ÷ 500 = **16 records/block**.

Total data blocks: 500 records ÷ 16 records/block = **32 blocks**.

Total index blocks: 32 data blocks ÷ 176 entries/block = **1 index block**.

So, **1 block** is required for the index.

Index 2+ Dense:

Index entry size: 200 bytes + 6 bytes = **206 bytes**.

Index entries per block: (8192 - 100) ÷ 206 = **39 entries/block**.

Total index blocks: 100 records ÷ 39 entries/block ≈ **3 index blocks**.

So, **3 blocks** are required for the index.

Index2 + Sparse:

We have 1 location per department. So, sparse index on department location isn't useful.

5. To hold the data file, we need n/3 blocks.

a) For Dense Index:

To hold the index file, we need n/10 blocks.

In total we need n/3 + n/10 = 13n/30 blocks

b) For Sparse Index:

To hold the index file, we need (n/3)/10 blocks.

In total we need n/3 + n/30 =11n/30 blocks.